La Scuola d'Italia / Dr. Huson / IB Math 12
3 November 2026

Nickname:

3.2 Classwork: Transformations of Trigonometric Functions

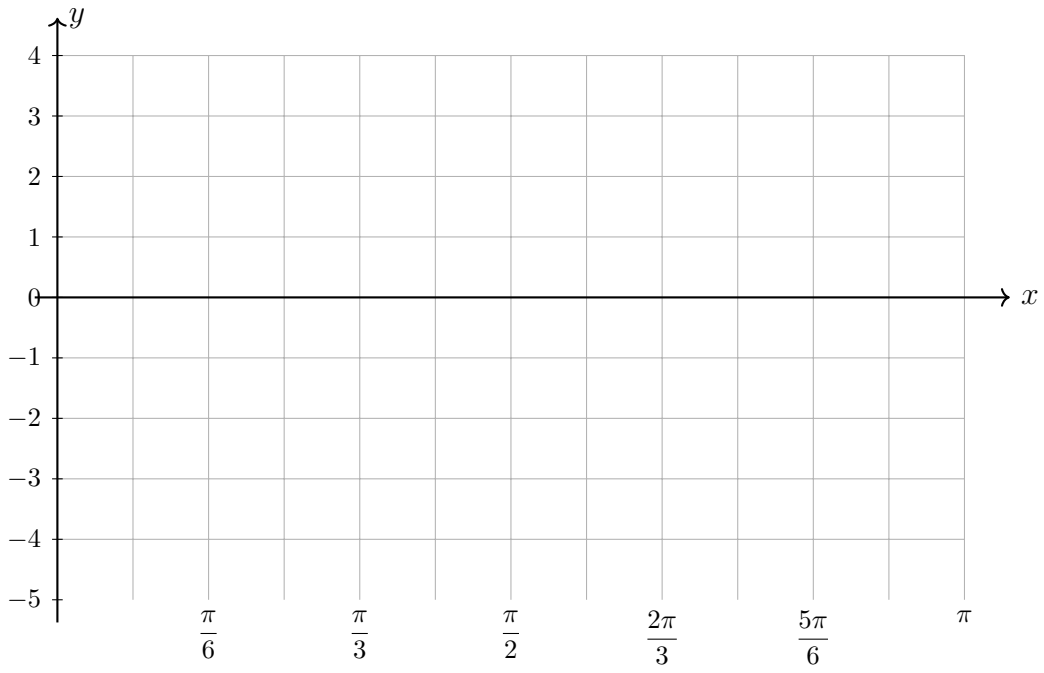
All angles are in radians. You may use your GDC.

1. The function f is defined by $f(x) = 4\sin\left(2\left(x - \frac{\pi}{6}\right)\right) - 1$.

   (a) Describe fully the sequence of transformations that maps the graph of $y = \sin x$ onto the graph of $y = f(x)$.

   (b) Write down the amplitude, the period and the range of f.

   (c) On the grid, sketch the graph of $y = f(x)$ for $0 \leq x \leq \pi$. Mark the maximum point and the minimum point, and write their *exact* coordinates.



2. The number of hours of daylight in New York, D, can be modelled by the function

$$D(t) = a\cos\big(b(t-c)\big) + d, \qquad 0 \leq t \leq 12,$$

where t is the time in months after 1 January, $a, b, c, d > 0$ and $c < 12$.

The longest day has 15.1 hours of daylight, at $t = 5.7$ (21 June). The shortest day has 9.3 hours of daylight, at $t = 11.7$ (21 December).

(a) Find the value of a and the value of d.

(b) Find the value of b and the value of c.

(c) Today is 3 November, $t = 10.1$. Use the model to find the number of hours of daylight today.

(d) Find the length of time during the year, in months, for which New York has more than 14 hours of daylight.